

VIVEK VYAS

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with master's level training in Artificial Intelligence and current studies in High Performance Computing and Quantum Computing. Practical experience in ML pipelines, computer vision, simulation based optimization, REST API integration, and data driven decision support for industrial use cases. Currently working as a Working Student Research Assistant at Fraunhofer FHR, with focus on radar AI, machine learning, and quantum computing. Comfortable with Python, SQL, REST APIs, Docker, AWS, and Azure, with focus on clean implementation, reproducibility, and data quality.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, CNN, LSTM, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Generative AI:** LLM Concepts, Prompt Engineering, Retrieval Augmented Generation, Embedding Based Retrieval
- **Computer Vision:** OpenCV, Image Classification, Object Detection, Image Segmentation, Anomaly Detection
- **MLOps and Deployment:** REST APIs, Flask, Docker, Kubernetes, Git, Containerized AI Services
- **Cloud and Infrastructure:** AWS, Microsoft Azure, Cloud based AI workflows
- **Data and Analytics:** Data Preprocessing, Data Pipelines, Exploratory Data Analysis, Error Analysis, Model Benchmarking
- **Simulation and Industrial AI:** Siemens Tecnomatix Plant Simulation, Production Optimization, Simulation Based ML Integration

PROFESSIONAL EXPERIENCE

Working Student Research Assistant

05/2026 – Present

Fraunhofer FHR, Wachtberg, Germany

- Supporting research work at the intersection of radar technology, machine learning, and quantum computing.
- Working on AI based analysis methods for intelligent radar applications.
- Applying Python based analysis and structured experimentation in a research environment.
- Contributing to research documentation, experiment tracking, and scientific computing workflows.

Master's Thesis – Münster, North Rhine-Westphalia, Germany

01/2025 – 10/2025

Topic: AI-Optimized Battery Manufacturing: Simulation & Smart Scheduling

Fraunhofer FFB, Münster, Germany

- Built ML based decision support logic for order scheduling using structured production data.
- Worked with simulation outputs to evaluate scheduling behavior and bottlenecks in a manufacturing flow.
- Developed preprocessing and feature engineering steps to improve model reliability and repeatability.
- Integrated model outputs through a Python based interface using REST APIs and Docker based workflows.

Trulloy Software and Solutions- Bhavnagar, Gujarat

05/2021 – 01/2022

Intern Position: Machine Learning Intern

- Supported development and testing of machine learning models using Python and TensorFlow.
- Prepared and cleaned structured datasets for model training and evaluation.
- Conducted experimental evaluation of ML models and documented results.
- Supported debugging and validation of data processing workflows.

EDUCATION

M.Sc. High Performance Computing / Quantum Computing

03/2026 - Ongoing

Deggendorf Institute of Technology (DIT) - Deggendorf

- Focus on parallel computing, quantum computing fundamentals, and advanced algorithm design.
- Study of supercomputing architectures, scalable system programming, and performance optimization.
- Relevant Coursework: Parallel Computing, Quantum Computing Fundamentals, Advanced Algorithms, High Performance Systems.

M.Sc. Artificial Intelligence for Smart Sensors and Actuators

10/2022 - 12/2025

Deggendorf Institute of Technology (DIT)- Technische Hochschule Deggendorf, Cham

- Focus on generative AI, deep learning, predictive modeling, industrial AI, and intelligent systems.
- Applied AI methods for data driven optimization, decision support, and system level integration.
- Relevant Coursework: Intelligent Systems, Autonomous Systems, Quantum Computing, Machine Learning

Bachelor's Degree: Electronics and Communication Engineering

07/2018 - 07/2022

Gujarat Technological University (GTU)- Government Engineering College, Bhavnagar

- Focus Areas: AI Algorithms, Control Systems, VLSI Design, IoT Protocols, Real Time Monitoring Systems

PROJECTS

Project 1: Building Age Prediction

- Developed convolutional neural network models to estimate building age from satellite images. Implemented preprocessing pipelines for large image datasets. Evaluated model performance using standard classification metrics.
- Technologies: Python, TensorFlow, OpenCV
- Link: <https://github.com/Vivek2960/Building-Age-Recognition>

Project 2: Face Mask Detection

- Developed a real time object detection system using convolutional neural networks. Implemented training and evaluation pipeline for image classification models. Optimized inference performance for real time deployment.
- Technologies: Python, TensorFlow, OpenCV
- Link: <https://github.com/Vivek2960/Face-MaskDetection>

Project 3: Stock Price Prediction Using LSTM

- Developed an LSTM based time series forecasting model for stock trend prediction. Exposed the trained model through a Flask REST API for integration into external applications. Evaluated forecasting performance using regression metrics and error analysis.
- Technologies: Python, TensorFlow, Keras, Flask
- Link: <https://github.com/Vivek2960/Stock-Prediction>

Project 4: NLP and Generative AI Systems

- Built NLP pipelines for text processing, information extraction, and AI assisted response generation. Developed LLM based applications using Python, prompt engineering, and retrieval based context handling. Integrated AI services through REST APIs and containerized workflows for reproducible deployment.
- Technologies: Python, LLMs, RAG, REST APIs, Docker

Project 5: Contour Detection Using Computer Vision

- Developed a computer vision system for object boundary detection using image segmentation techniques. Applied contour based analysis for pattern recognition and anomaly detection in image based inspection tasks. Processed visual input using OpenCV and evaluated segmentation output for detection quality.
- Technologies: Python, OpenCV, scikit-learn
- Link: <https://github.com/Vivek2960/Contour-Detection>

LANGUAGES

- German: A2 certified, currently learning B1 and improving
- English: C1

INTERESTS

Applied machine learning, radar AI, scientific computing, computer vision, simulation based optimization, high performance computing, industrial AI systems.